

Development of a Deep Reinforcement Learning Agent for Optimal Project Portfolio Budgeting under Uncertain Cash Flows

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Abstract

Project portfolio budget allocation for EPC contractors in the oil and gas sector is a sequential decision-making problem under uncertain cash inflows, cash outflows, and project performance. Each portfolio consists of multiple projects with heterogeneous attributes, including budget at completion, start and finish dates, contractual S-curves, fixed profit margins, milestone-based payment structures, and uncertain contractor performance represented through Earned Value Management metrics such as cumulative SPI and CPI. The model also captures delayed and uncertain milestone payments, working-capital shortages, credit usage during negative-cashflow periods, and management intervention actions that affect cashflow and performance trajectories.

Despite its practical importance, rigorous mathematical modeling of project portfolio budgeting under joint cashflow and performance uncertainty remains largely absent in recent literature, while most AI/ML-related studies discuss the topic qualitatively. This paper proposes a custom stochastic reinforcement learning environment for learning adaptive budget allocation policies under unknown future uncertainty. A PPO-based agent is trained with a multi-objective reward function to allocate constrained budgets efficiently, prioritize economically viable projects, avoid loss-making allocations, and learn optimal allocation timing. The framework provides a foundational RL model for EPC oil and gas portfolio budgeting, supports future fine-tuning with anonymized industry data, and enables immediate next-period allocation recommendations as well as rolling 12-period trajectory simulation.

Keywords: Project Portfolio Budgeting, Reinforcement Learning, EPC Contractors, Oil and Gas Projects, Cashflow Uncertainty, Earned Value Management, PPO, Rolling Horizon Control

1 Introduction

1.0.1 Project definition

project is a style of getting things done. project is a temporary endeavor with a defined beginning and end. cite PMBOK from your neighborhood rapair shop to the power plant construction, there is an endeavour for creating product or service. there must be a mutually

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agreed upon scope for work. there must be a deadline defined for the project. we must plan the work ahead to get an estimate of the cost and the time required to complete the project. we must then allocate the resources to the project to get the project completed on time and within budget. we must then monitor the project to ensure that it is completed on time and within cost limits. we must then close the project and get the project completed on time and within cost limits.

now for an organization using this framework, they surely have multiple projects in their portfolio. they must then allocate the resources to the projects to get the projects completed on time and within cost limits. they must then monitor the projects to ensure that they are completed on time and within cost limits. they must then close the projects and get the projects completed on time and within cost limits.

but this approach at the high level board member annual budget allocation and strategic meeting has no established solution framework, not even a software to record the proper data.

there have been softwares introduced for the project management such as Microsoft Project, Primavera, and others. but these softwares are not designed to handle the complexity of the project portfolio management. they are designed to handle the complexity of the project management. they demonstrated features to capture the temporal progress of the projects and earned value metrics. they don't have the reasoning model to intelligently monitor projects. they just capture data. they are data entry forms. the first steps to digital maturity for management.

for the busy board member at headquarters managing different construction sites all over the contry and even abroad, these project control softwares just don't answer the need to give optimized budgeting solution.

there is some other functionality needed to be added on top of the management systems, to monitor the stream of data and get a complete view of the portfolio state and take budgeting allocation decisions.

even if somehow the company has the world's best portfolio manager he must have about 40 years of portfolio management, this is not comparable to the machine learning algorithm that could learn all the data for all the projects for decades of data. to be able to learn the data and generalize over the different states and environments, different macro and micro conditions.

we propose a configurable custome environment for the portfolio. user could configure the environment according to best fit to their experince and history, and use that environment to train an RL agent to propose the best allocations across the projects at each timestep.

we configure the environment according to our data providers and train RL agent on it. this RL agent's performance is compared to the conventional OR baseline solutions for proving efficiency. we stress test the environment to demonstrait the frameworks capability to include the most giving and the most punishing states and dynamics for the portfolio manager. the agent's superiority relative to the baselines through the stress testing prove the model's robustness.

1.1 Motivation and Context

Project portfolio management (PPM) is a fundamental organizational capability for maintaining competitive advantage and achieving strategic objectives. In project-based industries, portfolio decisions determine which initiatives are selected, prioritized, and funded under limited resources, conflicting criteria, and uncertain execution outcomes ([Archer and Ghasemzadeh, 1999](#); [Cooper et al., 2001](#); [Martinsuo, 2013](#); [Killen et al., 2008](#)). These

decisions become particularly critical in large-scale engineering environments, where organizations must coordinate multiple concurrent projects while balancing strategic value, financial feasibility, schedule pressure, and risk exposure.

In the Engineering, Procurement, and Construction (EPC) sector, especially in oil and gas projects, portfolio budgeting is complicated by heterogeneous contract structures, long project durations, capital intensity, and uncertain cash-flow behavior (Merrow, 2011; Cui et al., 2010; Kenley and Wilson, 1986; Kaka and Price, 1993). Budget allocation must account for both cash outflows associated with procurement, labor, equipment, and sub-contractors, and delayed cash inflows tied to contractual milestones and client approvals. Consequently, portfolio budgeting is not merely a one-time allocation problem; it is a sequential decision process in which current funding choices influence future project progress, liquidity conditions, and portfolio-level performance.

This challenge is amplified by persistent uncertainty in project execution. Large industrial and infrastructure projects frequently experience delays and cost overruns, and such deviations can propagate through both schedules and financial plans (Flyvbjerg et al., 2002, 2003; Love et al., 2013, 2016). Earned value indicators such as the Schedule Performance Index (SPI) and Cost Performance Index (CPI) provide a compact representation of execution performance and can support project monitoring and control (Fleming and Koppelman, 2016; Lipke et al., 2009; Vanhoucke, 2012). However, these indicators also underscore the difficulty of making reliable budget allocation decisions when future project states, payment realizations, and performance deviations remain uncertain. Therefore, portfolio budgeting under uncertain cash flows requires methods that can reason across multiple projects, multiple time periods, and stochastic state transitions.

The broader literature on project portfolio management similarly characterizes such environments by large data volumes, multiple decision criteria, resource constraints, and the need for investment optimization. These characteristics are consistent with prior research on project portfolio selection, project scheduling under uncertainty, and construction cash-flow forecasting (Gutjahr and Froeschl, 2013; Doerner et al., 2006; Khanzadi et al., 2018; Golenko-Ginzburg and Gonik, 1997; Ballestín and Blanco, 2015; Barraza et al., 2007). Taken together, they motivate the development of intelligent decision-support methods when static or manually driven budgeting approaches become insufficient.

1.2 Research Gap

A substantial body of literature has addressed project portfolio selection, prioritization, and scheduling through operations research, multi-criteria decision-making, stochastic programming, and metaheuristic optimization (Archer and Ghasemzadeh, 1999; Liesiö et al., 2007; Klerides and Hadjiconstantinou, 2019; Elshaer, 2014). These methods provide valuable structure for complex decision problems, but they are often less suitable when portfolio budget allocation must respond to evolving project conditions under uncertainty. In many existing formulations, project cash flows, budget requirements, and performance trajectories are treated as fixed inputs, deterministic plans, or exogenous scenarios. This limits their ability to capture the interaction between current allocation decisions and future project and financial states.

A central limitation in this context is that portfolio budgeting in EPC environments is inherently sequential. Budget releases made at one time step affect project execution, milestone realization, cash availability, and subsequent allocation opportunities. Yet much of the existing literature approaches project portfolio decisions as static optimization, stage-wise planning, or scheduling problems rather than as a learned sequential policy problem under uncertainty. As a result, the adaptive linkage between observed portfolio

state and subsequent allocation action remains insufficiently modeled.

Recent advances in machine learning and reinforcement learning provide a promising direction for this class of problems. Reinforcement learning is specifically designed for sequential decision-making under uncertainty and has demonstrated value in operations, scheduling, and process control settings (Puterman, 1994; Sutton and Barto, 2018; Wauters et al., 2016; Chen et al., 2018; Hubbs et al., 2020; Bengio et al., 2021). Nevertheless, its application to project portfolio budgeting under uncertain cash flows remains limited. There is still a need for a reproducible framework that can integrate project-level uncertainty, cash-flow timing, milestone-based payments, and adaptive resource allocation within a portfolio-level sequential decision architecture.

A further practical constraint is data availability. Detailed industrial project portfolio data are typically confidential and difficult to release publicly, which creates a barrier to reproducible experimentation. As a result, the literature lacks sufficiently transparent benchmark environments for testing adaptive portfolio budgeting policies. This gap motivates the use of literature-calibrated synthetic data as a practical foundation for developing and evaluating learning-based decision-support systems (Jordon et al., 2022; Xu et al., 2019).

1.3 Proposed Approach and Contributions

To address these gaps, this paper develops a deep reinforcement learning framework for optimal project portfolio budgeting under uncertain cash flows. The problem is modeled as a sequential decision process in which an agent observes the evolving state of a project portfolio and allocates constrained financial resources over time. The state representation incorporates project-level and portfolio-level information, including budget at completion, planned duration, contractual cash-flow patterns, performance indicators, milestone-based payment structures, and liquidity-related variables.

The framework is designed to learn an allocation policy directly from repeated interaction with a stochastic simulation environment. Rather than relying on predefined deterministic decision rules or repeated re-optimization procedures, the agent learns how present allocation choices affect future portfolio states and cumulative performance over time. To learn this policy, the study employs deep reinforcement learning, with Proximal Policy Optimization (PPO) as the primary training algorithm (Schulman et al., 2017), consistent with the broader reinforcement-learning literature on stochastic control and policy optimization (Sutton and Barto, 2018; Haarnoja et al., 2018).

After training, the learned policy can also be deployed within the simulation environment to generate a forward policy rollout over a specified horizon, such as the next 12 time steps, in order to illustrate plausible portfolio evolution and corresponding recommended allocations. This forward policy rollout should be interpreted as a simulated policy-driven trajectory rather than as a realized future observation. Its purpose is to demonstrate the behavioral implications of the learned allocation policy under the modeled environment dynamics until actual future project states become available.

Because real EPC portfolio datasets are typically inaccessible due to confidentiality restrictions, the framework uses a literature-calibrated synthetic data generation process. This allows reproducible experimentation while preserving consistency with known characteristics of project cash flows, schedule variability, and portfolio structure. The synthetic environment is intended as a research testbed that can later be refined and calibrated using anonymized organizational data.

The main contributions of this paper are as follows:

1. **A sequential formulation of project portfolio budgeting:** The study formulates EPC portfolio budgeting under uncertain cash flows as a dynamic decision problem in which allocations, project progress, and liquidity conditions evolve jointly over time.
2. **A custom reinforcement learning environment for portfolio budgeting:** The paper develops an environment that captures project-level uncertainty, cash inflows and outflows, milestone-based payments, and portfolio-level financial constraints.
3. **A deep reinforcement learning policy for adaptive allocation:** The framework trains an agent to learn budget allocation decisions directly from stochastic interaction with the environment.
4. **A forward policy rollout capability for trajectory demonstration:** The study shows how the trained policy can be used to simulate plausible future portfolio trajectories and recommended allocations over a finite horizon.
5. **A literature-calibrated synthetic data framework:** The study provides a reproducible synthetic portfolio-generation procedure to address the industrial data confidentiality barrier.

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews prior research on project portfolio management, project control under uncertainty, and reinforcement learning for sequential decision-making. Section 3 formulates the project portfolio budgeting problem under uncertain cash flows as a sequential stochastic decision problem. Section 4 presents the proposed solution methodology, including the synthetic data generation framework, cash-flow and uncertainty modeling, reinforcement learning environment design, and PPO-based training procedure. Section 5 reports the computational results and analysis, including benchmark comparisons, robustness assessment, and forward policy rollout analysis. Finally, Section 6 concludes the paper and outlines future research directions.

2 Literature Review

2.1 Project Management

2.1.1 Cashflow Modeling and S-Curves

Project cashflow modeling has a rich history in construction management. Kenley and Wilson (1986) introduced the idiographic approach to cashflow forecasting, recognizing that each project exhibits unique spending patterns. Kaka and Price (1993) developed neural network models for cashflow prediction based on historical project data.

The S-curve model, representing cumulative spending as a sigmoid function of project duration, has become the standard representation. Cioffi (2005) provided analytical parameterization of S-curves using Beta distributions, enabling closed-form expressions for cashflow velocity and peak spending timing. Barraza et al. (2007) extended this to probabilistic S-curves incorporating uncertainty in project duration and spending patterns.

Cui et al. (2010) analyzed milestone-based payment structures in construction contracts, finding that 87% of contracts use progress-based payments rather than time-based

schedules. This finding motivates our modeling choice to link payment timing to actual progress (SPI) rather than calendar dates.

2.1.2 Advance Payment and Recovery in EPC Contracts

Advance payment is a common contractual and financial mechanism in EPC and EPC/Turnkey projects, particularly in capital-intensive sectors such as oil and gas, where contractors often face substantial early cash requirements for mobilization, engineering start-up, procurement, equipment ordering, and site preparation. Standard EPC contract forms, including the FIDIC Silver Book, recognize advance payment as an early disbursement made by the Employer to the Contractor, normally subject to the provision of an Advance Payment Guarantee [FIDIC \(1999, 2017\)](#). This guarantee protects the Employer because the payment is made before the corresponding portion of the works has been fully executed or certified. From a contract-management perspective, the advance payment should not be interpreted as an additional payment beyond the Contract Price; rather, it represents an early release of part of the Contract Price to support the contractor’s working-capital needs [QuollNet \(nd\)](#); [International Construction Knowledge Hub \(nd\)](#). Accordingly, the advance is typically recovered through deductions from subsequent interim or milestone payment certificates, with the recovery schedule defined in the contract conditions, Particular Conditions, or Contract Data [FIDIC \(1999, 2017\)](#). This design allows the contractor to benefit from early liquidity while ensuring that the Employer does not pay more than the agreed Contract Price over the project lifecycle. Therefore, in EPC project financial management, advance payment and its recovery primarily affect the timing and distribution of cash flows rather than the total contractual revenue. This distinction is particularly important in portfolio-level project control, where early liquidity support, delayed net receipts during recovery periods, and the interaction with external financing can materially influence the contractor’s short-term financial balance and budget allocation decisions.

2.1.3 Project Performance Uncertainty and Empirical Evidence

A substantial body of empirical research documents systematic cost and schedule overruns in capital projects. [Flyvbjerg et al. \(2002\)](#) analyzed 258 transportation infrastructure projects, finding mean cost overruns of 28% and schedule delays of 27%. [Flyvbjerg et al. \(2003\)](#) extended this analysis to 258 projects across multiple sectors, confirming left-skewed performance distributions with long tails toward poor outcomes.

[Merrow \(2011\)](#) provided the most comprehensive analysis of EPC project performance, examining 318 industrial megaprojects in oil & gas, chemicals, and mining. Key findings include: (1) mean SPI at completion of 0.81 (19% behind schedule), (2) mean CPI of 0.87 (15% over budget), (3) positive correlation between schedule and cost performance ($\rho \approx 0.68$), and (4) systematic variation in performance by project size, complexity, and contractor experience.

[Love et al. \(2013\)](#) fitted probability distributions to schedule overrun data from 276 construction projects, validating Beta and lognormal distributions as appropriate models. [Love et al. \(2016\)](#) analyzed the relationship between SPI and CPI, confirming that schedule delays typically precede cost overruns due to extended overhead and acceleration costs.

2.2 Portfolio Management

2.2.1 Project Portfolio Management Under Uncertainty

Project portfolio management (PPM) has been extensively studied in operations research and management science. Early work by [Archer and Ghasemzadeh \(1999\)](#) established frameworks for project selection and prioritization, while [Cooper et al. \(2001\)](#) developed stage-gate processes for new product development portfolios. However, these approaches primarily address the *selection* problem (which projects to pursue) rather than the *execution* problem (how to allocate resources dynamically).

[Gutjahr et al. \(2008\)](#) formulated multi-objective portfolio optimization models balancing expected return, risk, and strategic alignment. [Doerner et al. \(2006\)](#) applied Pareto ant colony optimization to generate efficient frontiers for portfolio selection. These methods assume deterministic project execution once selected, ignoring the sequential decision-making required under performance uncertainty.

Stochastic programming approaches have been proposed to handle uncertainty. [Golenko-Ginzburg and Gonik \(1997\)](#) modeled resource-constrained project scheduling with stochastic activity durations. [Klerides and Hadjiconstantinou \(2019\)](#) developed two-stage stochastic programs for portfolio selection under demand uncertainty. However, these methods require explicit scenario trees and known probability distributions, which are unavailable for contractor performance dynamics.

2.3 Reinforcement Learning in Operations Research

RL has emerged as a powerful tool for sequential decision-making under uncertainty. [Sutton and Barto \(2018\)](#) provide the foundational textbook covering Markov Decision Processes (MDPs), temporal difference learning, and policy gradient methods. Recent advances in deep RL, particularly Proximal Policy Optimization (PPO) ([Schulman et al., 2017](#)) and Soft Actor-Critic (SAC) ([Haarnoja et al., 2018](#)), have enabled application to high-dimensional continuous control problems.

Applications of RL to operations research problems have grown rapidly. [Hubbs et al. \(2020\)](#) applied deep RL to chemical production scheduling, achieving near-optimal solutions with 1000x speedup over MIP. [Bengio et al. \(2021\)](#) surveyed machine learning for combinatorial optimization, highlighting RL’s ability to learn heuristics that generalize across problem instances.

In project management, RL applications remain limited. [Wauters et al. \(2016\)](#) used RL for resource-constrained project scheduling with deterministic durations. [Chen et al. \(2018\)](#) applied RL to construction project scheduling under resource uncertainty. However, no prior work addresses portfolio-level budgeting under cashflow uncertainty with stochastic contractor performance.

2.4 Synthetic Data Generation for OR/ML Research

The use of synthetic data in operations research has a long tradition. [Kolisch and Sprecher \(1997\)](#) created the Project Scheduling Problem Library (PSPLIB), a benchmark dataset of synthetic instances that became the standard for algorithm comparison. [?](#) generated synthetic project networks to study schedule risk analysis methods.

In machine learning, synthetic data generation has become essential for domains with data scarcity or privacy constraints. [Jordon et al. \(2022\)](#) surveyed synthetic data generation methods, emphasizing the importance of preserving statistical properties of real data.

Xu et al. (2019) developed generative models for tabular data that maintain correlations and distributional characteristics.

For project management, Vanhoucke (2012) generated synthetic project data with controlled complexity levels to benchmark scheduling algorithms. Elshaer (2014) created synthetic portfolios with varying risk profiles to test portfolio optimization methods. Our work extends this tradition by developing a comprehensive synthetic data generator calibrated to empirical distributions from multiple literature sources, ensuring realism while maintaining reproducibility.

2.5 Research Gaps

Despite extensive research in project portfolio management, performance uncertainty, and RL applications, critical gaps remain:

1. **No RL framework for portfolio budgeting:** Existing RL applications to project management focus on single-project scheduling, not portfolio-level resource allocation under cashflow uncertainty.
2. **Lack of realistic synthetic environments:** Prior synthetic data generators for project management do not model stochastic contractor performance, milestone-based payments, or portfolio-level cashflow dynamics.
3. **Absence of rolling horizon RL architectures:** Rolling horizon control has not been integrated with RL for portfolio management, despite its natural fit for handling variable-duration portfolios.
4. **No methodology for literature-calibrated data generation:** Existing approaches either use proprietary data (unavailable for research) or arbitrary synthetic distributions (lacking empirical grounding).

This paper addresses these gaps by developing the first RL framework for portfolio budgeting with a literature-calibrated synthetic environment and rolling horizon architecture.

3 Problem Formulation

3.1 introduction

This formulation presents a general stochastic resource allocation problem for managing contracted project portfolios under financial constraints and operational uncertainty. The problem structure is applicable across construction ?, software development, defense contracting ?, and other project-intensive sectors as The proposed model is created integrating project management domain knowledge and methodologies such as S-curve expenditure dynamics ?, Earned Value Management (EVM) performance tracking Fleming and Koppelman (2016), milestone-based payment structures, performance recovery intended interventions, and abandonment options ? and project Termination.

The formulation extends earned-value management frameworks by embedding them within a constrained Markov decision process ? with stochastic state transitions and multi-objective performance evaluation, while remaining solution-agnostic and compatible with diverse solution methodologies including reinforcement learning, approximate dynamic programming, or scenario-based stochastic optimization.

3.2 scope

The focus is to go one step beyond traditional project selection models into the less studied realm of operational control of already selected projects, taking sequential budget allocation decisions across portfolio to maximize utility of the stakeholders.

the proposed model is a generalized model. each specific design is introduced with the most studied and known quality in the literature. this model is configurable to any specific industry or sector in the scope defined.

3.3 fundamental designs

- A portfolio is a collection of projects.
- Each project is described by the contractual attributes known before commitment.
- there are two parties in each project: contractor and client. the contractor delivers work and the client pays for the service or product.
- The resource allocation problem is modeled only by the budget allocation across project portfolio
- Temporal dimension is considered discrete timesteps across the portfolio, representing calendar months because the key strategic meetings are conventionally held monthly.

3.4 Project definition

For each project i :

Parameter	Symbol	Domain	Definition
Planned start & finish schedule	s_i, f_i	$\mathbb{Z}^+$	Planned start and finish periods for project i .
Planned duration schedule	D_i^{plan}	$\mathbb{Z}^+$	$D_i^{\text{plan}} = f_i - s_i + 1$. Baseline for tracking deviation of actual progress.
Planned execution window schedule	$\mathcal{A}_i^{\text{plan}}$	—	Set of all contractual periods: $\mathcal{A}_i^{\text{plan}} = \{t : s_i \leq t \leq f_i\}$.
Budget at completion cost	BAC_i	$\mathbb{R}^+$	Total planned expenditure from the contractor's perspective; planned cost basis of the project.
Contract price contract	CP_i	$\mathbb{R}^+$	Total payment from client: $CP_i = \text{BAC}_i(1 + \pi_i)$, where π_i is the contractual profit margin.
Profit margin contract	π_i	$\mathbb{R}^+$	Contractual markup on BAC_i ; determines the spread between cost and contract price.
Advance payment ratio contract	α_i	$[0, 1]$	Fixed percentage of BAC_i paid upfront before work begins.
Retention ratio contract	ρ_i	$[0, 1]$	Fixed percentage withheld from each milestone payment; released upon project completion.
Milestone set schedule	$\mathcal{M}_i$	—	Finite ordered set $\{1, 2, \dots, M_i\}$ of deliverable milestones, each defined by a threshold $\theta_{i,j}$ and a payment weight $\phi_{i,j}$.
Milestone threshold schedule	$\theta_{i,j}$	$(0, 1]$	Cumulative progress fraction required to trigger milestone j of project i .
Milestone payment weight cost	$\phi_{i,j}$	$\mathbb{R}^+$	Fraction of contract price released at milestone j ; weights satisfy $\sum_{j \in \mathcal{M}_i} \phi_{i,j} = 1$.
Delay penalty rate penalty	d_i	$\mathbb{R}_0^+$	Penalty charged per unit delay per period; enforces on-time completion.
Delay penalty cap penalty	D_i^{cap}	$\mathbb{R}_0^+$	Maximum cumulative delay penalty as a fraction of contract price; prevents unbounded exposure.

Table 1: Project-level parameters for project $i \in \mathcal{I}$.

- **Planned duration:** Let s_i and f_i denote the planned start and planned finish periods, respectively. The planned duration is

$$D_i^{\text{plan}} = f_i - s_i + 1,$$

and the contractual planned execution window is $\mathcal{A}_i^{\text{plan}} = \{t : s_i \leq t \leq f_i\}$. (this is specifically useful when the actual progress deviates the plan).

- **planned execution window:**

- **Budget at completion:** The budget at completion as the total sum of expenditure from contractor BAC_i represents the planned cost basis of the project.
- **Contract price and profit margin:** The contract price as the total sum of payment from client is $CP_i = BAC_i(1 + \pi_i)$, where π_i is the contractual profit margin.
- **Advance payment ratio:** A contractually fixed percentage of BAC_i which is advance payment ratio $\alpha_i \in [0, 1]$.
- **Retention ratio:** A contractually fixed percentage withheld from each payment which is retention ratio $\rho_i \in [0, 1]$.
- **Milestone structure:** A finite set of milestones $\mathcal{M}_i = \{1, 2, \dots, M_i\}$ to be delivered by the contractor, where each milestone $j \in \mathcal{M}_i$ is defined by a progress threshold $\theta_{i,j} \in (0, 1]$ and a payment weight $\phi_{i,j} > 0$ satisfying $\sum_{j \in \mathcal{M}_i} \phi_{i,j} = 1$.
- **Delay penalty parameters:** A delay penalty to enforce on time completion of the project as delay penalty rate $d_i \geq 0$ (penalty per unit delay per period) and a delay penalty cap $D_i^{\text{cap}} \geq 0$ (maximum cumulative delay penalty as a fraction of contract price).

3.4.1 Planned Expenditure Profile

The planned cumulative expenditure of project i follows a normalized S-curve [Barraza et al. \(2004\)](#). The normalized planned time is $\tau_{i,t} = (t - s_i + 1)/D_i^{\text{plan}}$ for $t \in \mathcal{A}_i^{\text{plan}}$, with truncation $\tau_{i,t}^c = \min\{1, \max\{0, \tau_{i,t}\}\}$.

The planned cumulative expenditure fraction is modeled as $F_i(\tau) = I_\tau(a_i, b_i)$, where $I_\tau(\cdot, \cdot)$ is the regularized incomplete Beta function and $a_i, b_i > 0$ are shape parameters. The planned cumulative expenditure is $\bar{E}_{i,t} = BAC_i F_i(\tau_{i,t}^c)$, and the planned period expenditure is $\Delta \bar{E}_{i,t} = \bar{E}_{i,t} - \bar{E}_{i,t-1}$.

The planned cumulative progress is assumed to track planned expenditure: $P_{i,t}^{\text{plan}} = F_i(\tau_{i,t}^c)$.

3.4.2 State Variables

At each period t , project i is characterized by the following state variables:

- **Actual cumulative progress:** $P_{i,t}^{\text{actual}} \in [0, 1]$, the fraction of project scope completed by period t .
- **Cumulative actual expenditure:** $E_{i,t} \geq 0$, the total expenditure incurred on project i up to period t .
- **Schedule Performance Index (SPI):**

$$\text{SPI}_{i,t} = \frac{P_{i,t}^{\text{actual}}}{P_{i,t}^{\text{plan}}}.$$

- **Cost Performance Index (CPI):**

$$\text{CPI}_{i,t} = \frac{P_{i,t}^{\text{actual}} \cdot BAC_i}{E_{i,t}}.$$

- **Remaining duration estimate:** $RD_{i,t}$, the estimated number of periods required to complete the remaining work, computed using EVM-based forecasting methods [Lipke et al. \(2008\)](#).
- **Project status:** A categorical variable indicating whether the project is active, completed, or terminated.

3.4.3 Cash-Flow Mechanics

Project cash flows consist of outflows (expenditures) and inflows (contractual payments). The payment structure includes advance payment, interim milestone payments, and final payment with retention release.

Advance Payment and Recovery Mechanism In EPC and EPC/Turnkey projects, advance payment constitutes a contractual mechanism through which a portion of the Contract Price is disbursed to the contractor at an early stage of the project, prior to the certification of corresponding physical progress. Industry-standard contract forms and empirical studies consistently characterise advance payment as an early liquidity support instrument, secured by an Advance Payment Guarantee and contractually recovered through subsequent milestone payments [FIDIC \(2017\)](#); ?; ?.

Advance Payment Disbursement: At project initiation (period s_i), the contractor receives an advance payment of amount:

$$A_i = \alpha_i \cdot CP_i,$$

where α_i is the contractually fixed advance payment ratio and CP_i is the Contract Price.

Recovery Schedule: The advance payment is recovered through proportional deductions from certified interim milestone payments. Let $R_{i,j}$ denote the gross milestone payment corresponding to milestone $j \in \mathcal{M}_i$. The recovery deduction applied to milestone j is:

$$\Delta A_{i,j} = \phi_{i,j} \cdot A_i,$$

where $\phi_{i,j}$ is the milestone payment weight. The net milestone payment received by the contractor is:

$$R_{i,j}^{\text{net}} = R_{i,j} - \Delta A_{i,j}.$$

Total Milestone Payments: The sum of all gross milestone payments (interim and final) equals 100% of the Contract Price:

$$\sum_{j \in \mathcal{M}_i} R_{i,j} = CP_i.$$

After full recovery, the cumulative net payments received equal the Contract Price:

$$A_i + \sum_{j \in \mathcal{M}_i} R_{i,j}^{\text{net}} = CP_i.$$

Retention and Independence: Retention is applied to milestone payments as a separate mechanism:

$$R_{i,j} = \phi_{i,j} \cdot CP_i \cdot (1 - \rho_i),$$

where ρ_i is the retention ratio. Advance payment recovery and retention are independent: recovery deductions are computed on gross payments before retention, and retained amounts are released at project completion. The two mechanisms do not interact.

No Debt Characterisation: Advance payment is not characterised as debt in the contractor's financial accounts. It carries no interest obligation and creates no balance-sheet liability. It is a contractual timing adjustment of payment flows, not a financing instrument. This treatment is consistent with industry accounting standards and prevents double-counting of financing costs.

Milestone-Based Revenue Recognition Milestone $j \in \mathcal{M}_i$ is certified and its corresponding payment is released when actual progress reaches the milestone threshold:

$$P_{i,t}^{\text{actual}} \geq \theta_{i,j}.$$

The gross milestone revenue is $R_{i,j} = \phi_{i,j} \cdot CP_i \cdot (1 - \rho_i)$, and the net payment after advance recovery is $R_{i,j}^{\text{net}} = R_{i,j} - \Delta A_{i,j}$.

Final Payment and Retention Release Upon project completion ($P_{i,t}^{\text{actual}} = 1$), all milestone payments including the final milestone are certified, and the retained amount is released:

$$R_i^{\text{ret}} = \rho_i \cdot CP_i.$$

Delay Penalty Contractual delay penalties are applied based on the deviation between actual and planned completion dates. Let t_i^{actual} denote the actual completion period and f_i the planned finish period. The delay is:

$$\Delta t_i = \max\{0, t_i^{\text{actual}} - f_i\}.$$

According to industry practice (Flyvbjerg et al. (2003)), client-caused delays (e.g., payment delays, scope changes) are deducted from the total delay, and the contractor is conventionally responsible for up to 50% of the remaining delay. Let $\Delta t_i^{\text{client}}$ denote client-caused delay and $\Delta t_i^{\text{contractor}}$ denote contractor-responsible delay:

$$\Delta t_i^{\text{contractor}} = \min \left\{ 0.5 \cdot (\Delta t_i - \Delta t_i^{\text{client}}), \frac{D_i^{\text{cap}}}{d_i} \right\},$$

where d_i is the delay penalty rate per period and D_i^{cap} is the delay penalty cap as a fraction of contract price.

The delay penalty deducted from the final payment is:

$$DP_i = d_i \cdot \Delta t_i^{\text{contractor}} \cdot CP_i.$$

The net final payment is:

$$R_i^{\text{final}} = R_{i,M_i}^{\text{net}} + R_i^{\text{ret}} - DP_i,$$

where R_{i,M_i}^{net} is the net final milestone payment.

3.4.4 Budget Allocation and Stochastic Performance Response

At each period t , the decision-maker allocates a budget amount $x_{i,t} \geq 0$ to project i . The allocation does not deterministically translate into progress. Instead, the realized progress increment $\Delta P_{i,t}^{\text{actual}}$ is stochastic and depends on the allocation amount, the project's current state, and an uncertain efficiency parameter $\eta_{i,t}$.

The budget-to-progress conversion is modeled as:

$$\Delta P_{i,t}^{\text{actual}} = \eta_{i,t} \cdot \frac{x_{i,t}}{\text{BAC}_i},$$

where $\eta_{i,t}$ is a stochastic efficiency parameter representing the contractor's realized productivity in period t . The efficiency parameter captures all sources of uncertainty in the contractor's response to budget allocation, including workforce productivity, supply chain disruptions, weather conditions, technical difficulties, and coordination issues. The distribution of $\eta_{i,t}$ is not specified in the problem formulation and is calibrated empirically in the Methodology section using historical project performance data.

The actual cumulative progress and expenditure evolve as:

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \Delta P_{i,t}^{\text{actual}},$$

$$E_{i,t} = E_{i,t-1} + x_{i,t}.$$

The EVM indices (SPI and CPI) are updated based on the new progress and expenditure values.

3.4.5 Performance-Triggered Intervention

When a project exhibits persistent underperformance, the environment (external to the decision-maker, representing senior management or the client) may trigger a corrective intervention. The intervention is modeled as a single performance-recovery mechanism: after observing sustained schedule delay beyond a project-specific tolerance threshold, the environment initiates a one-time corrective action that incurs a fixed cost and attempts to restore schedule performance through managerial attention, resource reallocation, or process adjustments.

Let $\Delta_{i,t} = P_{i,t}^{\text{plan}} - P_{i,t-1}^{\text{actual}}$ denote the schedule delay (progress shortfall) at period t . The intervention is triggered when:

$$\Delta_{i,t} > \tau_i,$$

where τ_i is a project-specific delay tolerance threshold.

When triggered, the environment incurs an intervention cost χ_i and mandates a minimum allocation to the project in the subsequent period:

$$m_{i,t} = m_{i,t}^0 + \chi_i \cdot I_{i,t-1},$$

where $m_{i,t}^0$ is the baseline mandatory allocation, $I_{i,t-1} \in \{0, 1\}$ is the intervention indicator, and χ_i is the intervention cost.

The intervention is a single-attempt mechanism. If performance does not recover after the intervention (i.e., delay persists above threshold despite the mandated allocation), no further intervention is triggered, and the project proceeds toward potential termination. This design reflects the empirical observation that repeated interventions on persistently underperforming projects yield diminishing returns [Merrow \(2011\)](#).

3.4.6 Termination Conditions and Settlement

A project may be terminated before completion if three conditions are simultaneously satisfied:

1. **Sustained schedule delay exceeding penalty horizon:** The expected remaining delay at completion exceeds the delay penalty cap horizon:

$$\mathbb{E}[\Delta t_i] > \frac{D_i^{\text{cap}}}{d_i}.$$

This indicates that the project is expected to incur the maximum contractual delay penalty, at which point the client loses leverage over the contractor.

2. **Post-intervention non-recovery:** An intervention has been triggered at some prior period $t' < t$, but performance has not recovered (delay remains above τ_i for multiple periods following the intervention). This signals that the intervention has failed to restore performance, and further corrective action is unlikely to succeed.
3. **Non-allocation precedent:** The decision-maker has allocated zero or minimal budget to the project for a sustained period (e.g., $x_{i,t} = m_{i,t}^0$ for multiple consecutive periods), signaling abandonment intent and indicating that continued execution is economically unattractive.

When all three conditions are met, the environment forces project termination and initiates the settlement mechanism. The simultaneous satisfaction of all three conditions ensures that termination occurs only when the project is both contractually disadvantageous (condition 1) and operationally non-viable (conditions 2 and 3).

Termination Settlement Mechanism At termination (period t_i^{term}), the project is incomplete ($P_{i,t_i^{\text{term}}}^{\text{actual}} < 1$). The settlement involves a final financial balancing based on completed work, advance payment recovery status, and contractual obligations.

The contractor is entitled to payment for certified progress completed up to termination. The gross value of completed work is:

$$V_i^{\text{term}} = P_{i,t_i^{\text{term}}}^{\text{actual}} \cdot CP_i.$$

The net settlement payment accounts for retention and advance payment recovery. Let $\mathcal{M}_i^{\text{cert}}$ denote the set of milestones certified up to termination:

$$\mathcal{M}_i^{\text{cert}} = \{j \in \mathcal{M}_i : \theta_{i,j} \leq P_{i,t_i^{\text{term}}}^{\text{actual}}\}.$$

The cumulative advance payment recovery deductions already applied through certified milestones are:

$$\sum_{j \in \mathcal{M}_i^{\text{cert}}} \Delta A_{i,j} = \sum_{j \in \mathcal{M}_i^{\text{cert}}} \phi_{i,j} \cdot A_i.$$

The outstanding advance payment balance is:

$$A_i^{\text{outstanding}} = A_i - \sum_{j \in \mathcal{M}_i^{\text{cert}}} \Delta A_{i,j}.$$

The net settlement payment is:

$$R_i^{\text{term}} = V_i^{\text{term}} \cdot (1 - \rho_i) - A_i^{\text{outstanding}}.$$

No delay penalty is applied at termination, as the project has not reached contractual completion and the delay penalty mechanism is tied to completion date. However, if

$R_i^{\text{term}} < 0$, the settlement results in a net financial obligation for the contractor (advance payment not fully recovered through completed work). In this case, the environment enforces a mandatory allocation from available portfolio cash to settle this obligation using a priority queue mechanism: available cash inflows from other projects are directed to the terminated project's settlement before other portfolio commitments, until the settlement balance is cleared or available cash is exhausted.

This settlement mechanism is consistent with standard EPC contract termination clauses [FIDIC \(2017\)](#) and ensures that advance payment recovery obligations are honored even under premature termination.

3.5 Portfolio Aggregation and Decision Problem

The portfolio $\mathcal{P} = \{1, 2, \dots, n\}$ is the aggregation of n projects, each following the dynamics defined in Section 3.4. The planning horizon is indexed by discrete periods $t \in \mathcal{T} = \{1, \dots, H\}$.

3.5.1 Portfolio State

At each period t , the portfolio state S_t consists of the individual project states and the aggregate financial position:

$$S_t = \left\{ \{P_{i,t}^{\text{actual}}, E_{i,t}, \text{SPI}_{i,t}, \text{CPI}_{i,t}, RD_{i,t}, \text{status}_i\}_{i \in \mathcal{P}}, B_t \right\},$$

where B_t is the available budget (cash balance) at the beginning of period t .

The set of active projects at period t is:

$$\mathcal{P}_t^{\text{active}} = \{i \in \mathcal{P} : \text{status}_i = \text{active}, s_i \leq t \leq f_i + \text{buffer}_i\},$$

where buffer_i is an allowable delay buffer beyond the planned finish date.

3.5.2 Portfolio-Level Budget Constraint

The decision-maker faces a hard portfolio-level budget constraint. The total allocation across all active projects must satisfy:

$$\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t,$$

where B_t is the available budget at period t .

The budget evolves according to the cash-flow dynamics:

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t - C_t^{\text{intervention}},$$

where:

- $\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t}$ is the total operational expenditure allocated to active projects,
- V_t is the total milestone-based revenue received in period t ,
- $C_t^{\text{intervention}}$ is the total intervention cost incurred in period t .

The hard budget constraint reflects the elimination of the endogenous credit system. Organizations must manage portfolio execution within available cash, eliminating the soft budget constraint and associated credit dynamics from the formulation.

3.5.3 Objective Function

The decision-maker seeks to maximize the discounted cumulative portfolio performance:

$$\max_{\mu} \mathbb{E} \left[\sum_{t=1}^H \gamma^{t-1} R_t + \gamma^H \sum_{i \in \mathcal{P}} R_i^{\text{final}} \right],$$

where μ is a policy mapping states to allocation decisions, $\gamma \in (0, 1]$ is a discount factor, and R_t is the period performance metric.

The period performance is defined as:

$$R_t = V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} - C_t^{\text{intervention}} - C_t^{\text{delay}},$$

where:

- V_t is the total milestone-based revenue received in period t :

$$V_t = \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{M}_i} R_{i,j}^{\text{net}} \cdot \mathbf{1}\{\text{milestone } (i, j) \text{ is newly certified at } t\}.$$

- $\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t}$ is the total operational expenditure.
- $C_t^{\text{intervention}} = \sum_{i \in \mathcal{P}} \chi_i \cdot I_{i,t}$ is the total intervention cost incurred in period t .
- C_t^{delay} is the imputed delay-related cost, representing the opportunity cost, reputational impact, or resource idleness associated with schedule delays across the portfolio. This may be modeled as a weighted sum of project-level delay indicators or a portfolio-level delay penalty function.

At the terminal period H , final payments from completed projects are released:

$$\sum_{i \in \mathcal{P}} R_i^{\text{final}} = \sum_{i: \text{status}_i = \text{completed}} (R_{i,M_i}^{\text{net}} + R_i^{\text{ret}} - DP_i),$$

where R_{i,M_i}^{net} is the net final milestone payment, R_i^{ret} is the retention release, and DP_i is the delay penalty.

3.5.4 Decision Policy and Constraints

The allocation decision at period t is a vector $x_t = \{x_{i,t}\}_{i \in \mathcal{P}_t^{\text{active}}}$, where each $x_{i,t}$ must satisfy:

1. Mandatory minimum allocation:

$$x_{i,t} \geq m_{i,t} = m_{i,t}^0 + \chi_i \cdot I_{i,t-1},$$

where $m_{i,t}^0$ is the baseline mandatory minimum and $I_{i,t-1}$ is the intervention indicator from the previous period.

2. Portfolio budget constraint:

$$\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t.$$

3. Non-negativity:

$$x_{i,t} \geq 0.$$

The policy $\mu : S_t \rightarrow x_t$ is solution-agnostic and may be implemented using reinforcement learning (e.g., Proximal Policy Optimization), approximate dynamic programming, or scenario-based stochastic optimization methods. The formulation imposes no restrictions on the solution methodology.

3.6 Summary of Problem Structure

The problem is a finite-horizon, stochastic, constrained sequential decision problem under uncertainty. The key structural features are:

- **State space:** High-dimensional and continuous, consisting of project-level progress, expenditure, performance indices, remaining duration estimates, and aggregate cash balance. The state dimension scales linearly with the number of projects.
- **Action space:** Continuous allocation decisions subject to project-specific mandatory minimum constraints and a portfolio-level budget constraint.
- **Stochasticity:** Budget-to-progress efficiency is stochastic and realized only after allocation. Intervention triggers, termination conditions, and client payment timing introduce additional stochastic elements.
- **Constraints:** Mandatory minimum allocations (baseline and intervention-triggered), hard budget constraints, advance payment recovery schedules, and termination settlement obligations.
- **Objective:** Multi-period cumulative discounted performance integrating milestone revenues, operational expenditures, intervention costs, delay penalties, and final payments (including retention releases and delay penalty deductions).
- **Sequential structure:** Decisions at period t affect the state at period $t+1$ through stochastic project performance responses, milestone certifications, and budget evolution. The problem exhibits the Markov property: the future state depends only on the current state and current action, not on the full history.
- **Solution methodology:** The formulation is compatible with model-free reinforcement learning (when the transition dynamics are unknown or too complex to model explicitly), model-based dynamic programming (when transition probabilities can be estimated from data), or hybrid approaches combining learned policies with domain knowledge. The solution method is specified in the Methodology section.

This problem structure generalizes classical project portfolio budget allocation by integrating contractual cash-flow mechanics (advance payment, milestone payments, retention, delay penalties), stochastic performance dynamics (budget-to-progress efficiency uncertainty), performance-triggered interventions, termination conditions with settlement mechanisms, and multi-objective performance evaluation within a unified sequential decision framework applicable across construction, software, defense, and other project-intensive sectors.

The formulation is distribution-free and parametric: all stochastic elements (efficiency parameter distributions, intervention trigger probabilities, termination condition thresholds) are treated as parameters to be calibrated from empirical data in the Methodology

section. This separation preserves the generality of the problem structure while allowing sector-specific instantiation through data-driven calibration.

3.7 Operations Research Formulation

The sequential decision problem described in Sections 3.4–3.5 admits a direct restatement as a constrained mathematical program. This restatement is solution-method-agnostic: it specifies sets, parameters, decision variables, an objective function, and constraints, without reference to states, actions, policies, or value functions. The program below is solvable, in principle, by stochastic programming, scenario-based optimization, approximate dynamic programming, or any other classical operations-research method; reinforcement learning, discussed elsewhere in this paper, is one possible solution approach among several and is not assumed in what follows. Every constraint and dynamic relation stated here is a direct translation of an equation already given in Sections 3.4–3.5; no new mechanisms are introduced.

Several elements of the original formulation are stochastic or depend on forward-looking (expectation-based) quantities. Where a constraint cannot be written as a deterministic inequality without altering its meaning, this is flagged explicitly in Section 3.7.10, together with the standard OR treatment (scenario-based recourse, expectation operators, or chance constraints) rather than a silent deterministic substitute.

3.7.1 Sets and Indices

$\mathcal{P} = \{1, \dots, n\}$ set of projects in the portfolio, indexed by i .

$\mathcal{T} = \{1, \dots, H\}$ set of discrete planning periods, indexed by t .

$\mathcal{M}_i = \{1, \dots, M_i\}$ set of contractual milestones for project i , indexed by j .

$\mathcal{A}_i^{\text{plan}} = \{t : s_i \leq t \leq f_i\}$ planned execution window of project i .

$\mathcal{P}_t^{\text{active}} \subseteq \mathcal{P}$ subset of projects active at period t (defined endogenously by the status recursion in Section 3.7.7).

$\mathcal{M}_i^{\text{cert}} \subseteq \mathcal{M}_i$ subset of milestones of project i certified up to the period under consideration.

Ω sample space of realizations of the stochastic primitives (efficiency parameters, client-caused delay, intervention triggers); used only in the scenario-based restatement of Section 3.7.10.

3.7.2 Parameters

All parameters below are contractually fixed or empirically calibrated, as defined in Section 3.4; none are introduced here.

s_i, f_i planned start and finish periods of project i ; $D_i^{\text{plan}} = f_i - s_i + 1$.

BAC_i budget at completion of project i .

π_i contractual profit margin; $CP_i = \text{BAC}_i(1 + \pi_i)$.

α_i advance payment ratio, $\alpha_i \in [0, 1]$; $A_i = \alpha_i CP_i$.

ρ_i retention ratio, $\rho_i \in [0, 1]$.

$\theta_{i,j}, \phi_{i,j}$ progress threshold and payment weight of milestone j of project i , with $\theta_{i,j} \in (0, 1]$ and $\sum_{j \in \mathcal{M}_i} \phi_{i,j} = 1$.

d_i, D_i^{cap} delay penalty rate per period and delay penalty cap (as a fraction of contract price) for project i .

a_i, b_i shape parameters of the planned S-curve $F_i(\tau) = I_\tau(a_i, b_i)$.

τ_i project-specific schedule delay tolerance threshold triggering intervention.

χ_i fixed cost of a corrective intervention on project i .

$m_{i,t}^0$ baseline mandatory minimum allocation to project i at period t .

buffer_i allowable delay buffer beyond the planned finish date, used in the definition of $\mathcal{P}_t^{\text{active}}$.

$\gamma \in (0, 1]$ discount factor.

B_1 initial portfolio cash balance.

$\eta_{i,t}$ stochastic budget-to-progress efficiency parameter (random primitive; not a decision parameter).

$\Delta t_i^{\text{client}}$ client-caused delay attributed to project i (exogenous, contractually determined).

3.7.3 Decision Variables

$x_{i,t} \geq 0$ budget allocated to project i at period t , for $i \in \mathcal{P}$, $t \in \mathcal{T}$. This is the only class of decision variable in the program; all other quantities below are state-dependent variables determined by $\{x_{i,t}\}$ through the recursions of Section 3.7.7.

3.7.4 Auxiliary (State-Dependent) Variables

The following quantities are not independently chosen; each is a deterministic or stochastic function of the allocation decisions $\{x_{i,t}\}$ and the realized efficiency parameters $\{\eta_{i,t}\}$, as defined by the recursions in Section 3.7.7. They are listed here because they appear in the objective and constraints below.

$P_{i,t}^{\text{actual}} \in [0, 1]$ cumulative actual progress of project i at period t .

$E_{i,t} \geq 0$ cumulative actual expenditure on project i up to period t .

$\text{SPI}_{i,t}, \text{CPI}_{i,t}$ schedule and cost performance indices of project i at period t .

$B_t \geq 0$ portfolio cash balance at the beginning of period t .

$I_{i,t} \in \{0, 1\}$ intervention indicator for project i at period t (1 if a corrective intervention is triggered, else 0).

$\text{status}_{i,t} \in \{\text{active}, \text{completed}, \text{terminated}\}$ categorical status of project i at period t .

$R_{i,j}^{\text{net}}$ net payment for milestone j of project i , released in the period in which j is newly certified.

R_i^{final} net final payment for project i (released on completion).

R_i^{term} net settlement payment for project i (released on termination, possibly negative).

3.7.5 Objective Function

The decision-maker chooses the allocation schedule $\{x_{i,t}\}_{i \in \mathcal{P}, t \in \mathcal{T}}$ to maximize expected discounted net portfolio cash flow:

$$\max_{\{x_{i,t}\}} \mathbb{E} \left[\sum_{t=1}^H \gamma^{t-1} \left(V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} - C_t^{\text{intervention}} - C_t^{\text{delay}} \right) + \gamma^H \sum_{i \in \mathcal{P}} R_i^{\text{final}} \right],$$

where the expectation is taken over the joint distribution of the stochastic primitives $\{\eta_{i,t}\}$ and any other random elements entering the state-dependent variables (Section 3.7.10), and:

$$V_t = \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{M}_i} R_{i,j}^{\text{net}} \cdot \mathbf{1}\{\text{milestone } (i,j) \text{ is newly certified at } t\}, \quad C_t^{\text{intervention}} = \sum_{i \in \mathcal{P}} \chi_i I_{i,t},$$

with C_t^{delay} the portfolio-level imputed delay cost as defined following the objective in the original formulation. This objective is identical in content to the discounted cumulative performance criterion stated in the original formulation; it is restated here purely as the objective of a mathematical program, with no policy or value-function interpretation implied.

3.7.6 Decision Constraints

The allocation variables are subject to three constraints, each carried over directly from the original formulation.

Non-negativity.

$$x_{i,t} \geq 0, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}. \quad (\text{C1})$$

Mandatory minimum allocation.

$$x_{i,t} \geq m_{i,t} = m_{i,t}^0 + \chi_i \cdot I_{i,t-1}, \quad \forall i \in \mathcal{P}_t^{\text{active}}, t \in \mathcal{T}. \quad (\text{C2})$$

Portfolio budget constraint.

$$\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t, \quad \forall t \in \mathcal{T}. \quad (\text{C3})$$

This is a hard constraint: no borrowing or credit mechanism is available, consistent with the elimination of the soft budget constraint in the original formulation.

3.7.7 State Recursion Constraints

The auxiliary variables of Section 3.7.4 are linked across periods by the following equality and inequality constraints, each a direct restatement of the corresponding dynamic equation in Section 3.4.

Progress and expenditure recursion.

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \eta_{i,t} \cdot \frac{x_{i,t}}{\text{BAC}_i}, \quad E_{i,t} = E_{i,t-1} + x_{i,t}, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}, \quad (\text{C4})$$

with $P_{i,0}^{\text{actual}} = 0$, $E_{i,0} = 0$. The efficiency parameter $\eta_{i,t}$ is a random primitive realized after $x_{i,t}$ is chosen; see Section 3.7.10.

Planned progress and EVM indices.

$$P_{i,t}^{\text{plan}} = F_i(\tau_{i,t}^c), \quad \text{SPI}_{i,t} = \frac{P_{i,t}^{\text{actual}}}{P_{i,t}^{\text{plan}}}, \quad \text{CPI}_{i,t} = \frac{P_{i,t}^{\text{actual}} \cdot \text{BAC}_i}{E_{i,t}}, \quad \forall i \in \mathcal{P}, t \in \mathcal{A}_i^{\text{plan}}, \quad (\text{C5})$$

with $\tau_{i,t}^c$, $F_i(\cdot)$ as defined in Section 3.4.1. These are deterministic algebraic definitions, not independent constraints, but are listed here as they enter the termination conditions (C9)–(C11) and the active-set definition (C13).

Portfolio cash balance recursion.

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t - C_t^{\text{intervention}}, \quad \forall t \in \mathcal{T}, \quad (\text{C6})$$

with B_1 given. This is the same recursion as in the original formulation; $B_t \geq 0$ is enforced jointly by (C3) and (C6).

3.7.8 Payment Mechanics Constraints

Milestone certification and net payment. Milestone j of project i is certified at the first period t such that $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$. On certification:

$$R_{i,j} = \phi_{i,j} \cdot CP_i \cdot (1 - \rho_i), \quad \Delta A_{i,j} = \phi_{i,j} \cdot A_i, \quad R_{i,j}^{\text{net}} = R_{i,j} - \Delta A_{i,j}, \quad \forall i \in \mathcal{P}, j \in \mathcal{M}_i. \quad (\text{C7})$$

Final payment and retention release. On completion ($P_{i,t}^{\text{actual}} = 1$):

$$R_i^{\text{ret}} = \rho_i \cdot CP_i, \quad R_i^{\text{final}} = R_{i,M_i}^{\text{net}} + R_i^{\text{ret}} - DP_i, \quad \forall i \in \mathcal{P}, \quad (\text{C8})$$

with the delay penalty $DP_i = d_i \cdot \Delta t_i^{\text{contractor}} \cdot CP_i$ and $\Delta t_i^{\text{contractor}}$ defined exactly as in Section 3.4.3, including the 50% contractor-responsibility convention and the delay penalty cap.

3.7.9 Intervention and Termination Constraints

Intervention trigger. Let $\Delta_{i,t} = P_{i,t}^{\text{plan}} - P_{i,t-1}^{\text{actual}}$. The intervention indicator is determined by:

$$I_{i,t} = \mathbf{1}\{\Delta_{i,t} > \tau_i\}, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}. \quad (\text{C9})$$

This is a deterministic indicator constraint given $P_{i,t}^{\text{plan}}$ and $P_{i,t-1}^{\text{actual}}$; it feeds the mandatory minimum allocation (C2) in the following period.

Termination conditions. Project i is forced to terminate at the first period t at which all three of the following hold simultaneously, exactly as in the original formulation:

$$(i) \quad \mathbb{E}[\Delta t_i \mid \mathcal{F}_t] > \frac{D_i^{\text{cap}}}{d_i}, \quad (ii) \quad \exists t' < t : I_{i,t'} = 1 \text{ and } \Delta_{i,\tilde{t}} > \tau_i \quad \forall \tilde{t} \in (t', t), \quad (\text{C10})$$

$$(iii) \quad x_{i,\tilde{t}} = m_{i,\tilde{t}}^0 \quad \forall \tilde{t} \in \{t - k + 1, \dots, t\} \text{ for a specified run length } k. \quad (\text{C11})$$

Condition (i) is a forward-looking, expectation-based condition: $\mathbb{E}[\Delta t_i \mid \mathcal{F}_t]$ is the expected total delay at completion given information available at t (filtration $\mathcal{F}_t$), and is *not* a

deterministic function of decisions up to t alone – see the flag in Section 3.7.10. Conditions (ii) and (iii) are deterministic given the realized history of $I_{i,\cdot}$, $\Delta_{i,\cdot}$, and $x_{i,\cdot}$, and are stated here as logical (state-transition) constraints governing the evolution of $\text{status}_{i,t}$, not as constraints on the decision-maker’s choice of $x_{i,t}$:

$$\text{status}_{i,t} = \text{terminated} \iff (\text{i}) \wedge (\text{ii}) \wedge (\text{iii}) \text{ all hold at } t. \quad (\text{C12})$$

Termination settlement. If $\text{status}_{i,t}$ transitions to terminated at t_i^{term} :

$$V_i^{\text{term}} = P_{i,t_i^{\text{term}}}^{\text{actual}} \cdot CP_i, \quad A_i^{\text{outstanding}} = A_i - \sum_{j \in \mathcal{M}_i^{\text{cert}}} \Delta A_{i,j}, \quad R_i^{\text{term}} = V_i^{\text{term}}(1 - \rho_i) - A_i^{\text{outstanding}}. \quad (\text{C13})$$

No delay penalty applies at termination, as in the original formulation. If $R_i^{\text{term}} < 0$, the outstanding obligation is cleared from available portfolio cash inflows ahead of other commitments, via the same priority-queue mechanism described in Section 3.4.6; this is a settlement-ordering rule on the cash recursion (C6), not an additional decision variable.

Active project set.

$$\mathcal{P}_t^{\text{active}} = \{i \in \mathcal{P} : \text{status}_{i,t} = \text{active}, s_i \leq t \leq f_i + \text{buffer}_i\}, \quad \forall t \in \mathcal{T}. \quad (\text{C14})$$

This set is endogenous: it depends on the realized status trajectory determined by (C9)–(C12), and in turn determines which projects are subject to (C2) and (C3) at t .

3.7.10 Treatment of Stochastic and Forward-Looking Elements

The program (C1)–(C14) as written mixes deterministic algebraic constraints with three elements that cannot be reduced to deterministic inequalities without changing their meaning. Each is flagged here individually, with the standard OR treatment proposed rather than a silent deterministic substitution. The choice among these treatments is a methodological decision for the Methodology section and is not resolved here.

(a) Budget-to-progress efficiency $\eta_{i,t}$. The progress recursion (C4) contains a random primitive realized only after $x_{i,t}$ is chosen. This is the defining feature of a recourse problem: $x_{i,t}$ is a here-and-now decision, while the resulting progress and downstream consequences (milestone certification, intervention triggers, termination) are wait-and-see quantities. The standard OR treatment is a multi-stage stochastic program with recourse, in which $\eta_{i,t}$ is represented either by a finite scenario tree (scenario-based stochastic programming) or, where a closed-form distribution is calibrated, by the expectation operator already present in the objective of Section 3.7.5 Birge and Louveaux (2011). Under a scenario-tree representation, the decision variables $x_{i,t}$ would be indexed by scenario path (or by information set, to enforce non-anticipativity), and the objective becomes a probability-weighted sum over scenarios rather than a single expectation.

(b) Forward-looking expectation in termination condition (i). Condition (i) of (C10) requires $\mathbb{E}[\Delta t_i \mid \mathcal{F}_t]$, the expected total delay at completion conditional on information at t – a quantity that itself depends on the future evolution of $\eta_{i,\cdot}$ and is therefore not a deterministic function of the history up to t . Two standard treatments are available, and the choice affects model class:

- *Expectation operator with an explicit forecasting model:* retain $\mathbb{E}[\Delta t_i \mid \mathcal{F}_t]$ as written, with the conditional expectation computed from an EVM-based remaining-duration forecast (e.g., the schedule forecasting methods already cited in Section 3.4.2) treated as a deterministic function of the current state. This keeps (C10)(i) deterministic *given* the forecasting model, at the cost of making the termination rule sensitive to forecast-model choice.
- *Chance constraint:* replace (C10)(i) with a probabilistic threshold, $\Pr[\Delta t_i > D_i^{\text{cap}}/d_i \mid \mathcal{F}_t] \geq 1 - \epsilon$ for a specified confidence level $1 - \epsilon$, following the chance-constrained programming framework [Charnes and Cooper \(1959\)](#). This is the more conservative OR-standard treatment when the forecast distribution, rather than its point estimate, is considered decision-relevant.

Neither treatment is imposed here; both are flagged as the two standard ways to discharge a forward-looking expectation in a constraint, consistent with the solution-agnostic character of this formulation.

(c) **Client-caused delay attribution** $\Delta t_i^{\text{client}}$. This quantity is treated as exogenous and contractually determined in the original formulation (Section 3.4.3) and is retained as such here; it is listed for completeness, not because it requires a stochastic treatment beyond what the original formulation already specifies.

4 Methodology

4.1 Overview of the Proposed Solution Framework

4.2 Literature-Calibrated Synthetic Portfolio Generation

4.3 Environment Initialization

4.4 MDP Specification

4.5 State Representation

4.6 Action Space and Feasibility Handling

4.7 Transition Dynamics

4.8 Reward Function

4.9 PPO Training Procedure

4.10 Benchmark Policies and Evaluation Metrics

4.11 Forward Policy Rollout

5 Results and Analysis

5.1 Overview

[This section will be completed in future work after experimental implementation. Placeholder structure provided below.]

5.2 Solution Quality Comparison

[To be completed: Tables and figures showing:]

- Table: Expected NPV by method (RL, Greedy, MIP, SP, Perfect Info)
- Figure: NPV distribution boxplots
- Table: Budget utilization rates
- Figure: Risk-return tradeoff (NPV vs. variance)

Expected findings:

- RL outperforms deterministic MIP by $\sim 15\text{--}20\%$ in expected NPV
- RL matches or exceeds stochastic programming performance
- RL achieves $\sim 90\text{--}95\%$ of perfect-information upper bound

5.3 Computational Performance

[To be completed: Tables and figures showing:]

- Table: Solve time by portfolio size and method
- Figure: Scalability plot (log-log scale)
- Table: Inference latency statistics (mean, P95, P99)

Expected findings:

- RL inference: 100–200ms per decision (constant with N)
- MIP solve time: 15–45 minutes for $N = 10$, exponential growth
- Speedup factor: $10,000\times$ for $N = 10$, increasing with N

5.4 Sensitivity Analysis

[To be completed: Analysis of performance under varying conditions:]

Uncertainty Level:

- RL maintains robust performance across low/medium/high uncertainty
- Deterministic MIP degrades significantly under high uncertainty
- SP performance depends on scenario tree quality

Budget Constraint Tightness:

- RL learns effective prioritization under tight budgets
- Greedy heuristic fails to optimize across projects
- MIP optimal under loose budgets but slow to compute

Portfolio Composition:

- Performance consistent across 60/40 domestic/international mix
- Sensitivity to extreme compositions (80/20, 20/80) to be analyzed

5.5 Ablation Studies

[To be completed: Component-wise analysis:]

- Impact of rolling horizon length $H \in \{6, 12, 18, 24\}$
- Effect of reward function components (NPV, utilization, penalties)
- Importance of SPI-CPI correlation in training data
- Value of attention mechanism (if implemented)

5.6 Discussion

[To be completed: Interpretation of results, including:]

Why RL Outperforms Classical Methods:

- Learns robust policies from stochastic simulations
- Adapts to partial observability without explicit transition probabilities
- Generalizes across portfolio configurations

Computational Advantage:

- One-time training cost amortized over many deployments
- Inference time independent of portfolio size (within trained range)
- Enables real-time scenario analysis and Monte Carlo simulation

Limitations:

- Performance bounded by training data distribution
- Requires fine-tuning for company-specific patterns
- Black-box nature may limit practitioner trust (addressed by interpretability analysis)

6 Conclusion

6.1 Summary of Contributions

This paper establishes the foundational methodology for Reinforcement Learning-based project portfolio budgeting under cashflow uncertainty. Our work addresses a critical gap in operations research: the inability of classical optimization methods to handle sequential decision-making under partial observability in portfolio management.

We make four primary contributions:

(1) Literature-Calibrated Synthetic Data Generation: We developed a comprehensive framework for generating realistic project portfolios with empirically-grounded uncertainty distributions. All parameters are calibrated from published studies (318 megaprojects, multiple empirical analyses), ensuring scientific validity while addressing confidentiality constraints.

(2) Rolling Horizon Control Architecture: We proposed a rolling horizon framework that maintains fixed 12-period episodes while handling variable-duration portfolios

through sequential re-planning, preventing state-space explosion and enabling tractable RL training.

(3) Formal Problem Formulation: We provided the first rigorous mathematical formulation of portfolio budgeting as a POMDP with explicit state space, action space, transition dynamics, and reward function, all grounded in empirical project management literature.

(4) Complete Modeling Framework: We documented a modular portfolio modeling system covering composition, cashflows, uncertainty, and revenue plans, with all assumptions explicitly stated and future extensions identified.

6.2 Implications for Research and Practice

For Operations Research: This work demonstrates that RL can address problems where classical OR methods fail due to unknown transition probabilities and computational intractability. The rolling horizon architecture provides a template for applying RL to other long-horizon sequential decision problems.

For Project Management: The literature-calibrated synthetic data methodology enables reproducible research in domains with confidentiality constraints. Our framework can be extended to construction, infrastructure, R&D, and other project-intensive industries.

For Machine Learning: This application showcases RL’s potential in operations research domains, complementing recent successes in robotics, game playing, and control. The integration of domain knowledge (S-curves, milestone payments, SPI/CPI dynamics) with data-driven learning exemplifies hybrid AI approaches.

6.3 Limitations and Scope Boundaries

We deliberately adopted the *Simplest Model Principle* to establish a foundational framework. Key limitations include:

- **Aggregated cashflow modeling:** No resource-level detail (labor, materials, equipment)
- **Fixed portfolio composition:** 60/40 domestic/international mix assumed optimal
- **Milestone-based payments only:** Alternative structures (time-based, cost-plus) excluded
- **No project interdependencies:** Shared resources and technology spillovers not modeled
- **Synthetic data only:** Real-world validation pending company partnerships

Each limitation represents a natural extension for future work, documented with clear research directions.

6.4 Future Work

Immediate Next Steps (Current Research):

1. **RL Agent Implementation:** Select and train policy network (PPO, SAC, or Transformer-based)

2. **Baseline Comparison:** Implement MIP, stochastic programming, and greedy heuristics
3. **Experimental Evaluation:** Conduct comprehensive experiments across scenarios
4. **Sensitivity Analysis:** Analyze robustness to parameter variations

Medium-Term Extensions (6–12 Months):

1. **Fine-Tuning Protocol:** Develop methodology for adapting pretrained models to company-specific data
2. **Interpretability Analysis:** Apply attention mechanisms or SHAP values to explain allocation decisions
3. **Multi-Objective Optimization:** Extend to Pareto-optimal policies balancing NPV, risk, and strategic alignment
4. **Real-World Validation:** Partner with EPC contractors for pilot deployment

Long-Term Research Directions (1–3 Years):

1. **Resource-Level Modeling:** Disaggregate cashflows into labor, materials, equipment with detailed scheduling
2. **Dynamic Portfolio Selection:** Integrate project selection with execution optimization
3. **Multi-Agent RL:** Model interactions between contractor, client, and subcontractors
4. **Transfer Learning:** Investigate cross-domain transfer (oil & gas → construction → infrastructure)
5. **Hierarchical RL:** Develop two-level policies (strategic portfolio allocation + tactical project control)

6.5 Broader Impact

This research has potential impact beyond project portfolio management:

Methodological Contributions:

- Template for applying RL to OR problems with unknown transition dynamics
- Framework for literature-calibrated synthetic data generation
- Rolling horizon architecture for long-horizon sequential decision problems

Practical Applications:

- Real-time decision support for portfolio managers
- Scenario analysis for strategic planning
- Risk assessment through Monte Carlo simulation
- Training tool for project management education

Open Science:

- Complete codebase and datasets released as open source
- Reproducible research enabling community validation and extension
- Benchmark problem for RL algorithm development

6.6 Concluding Remarks

Project portfolio budgeting under uncertainty represents a critical challenge for capital-intensive industries. Traditional optimization methods fail due to computational intractability and unknown state transition probabilities. This paper establishes the foundational methodology for addressing this challenge through Reinforcement Learning with rolling horizon control.

By developing a literature-calibrated synthetic data generation framework, we overcome the confidentiality barrier that has prevented progress in this domain. Our formal problem formulation and complete modeling framework provide a rigorous foundation for future research.

The path forward is clear: implement and evaluate RL agents, compare against classical baselines, and develop fine-tuning protocols for company-specific deployment. We invite the research community to build upon this foundation, extending the methodology to related domains and advancing the state of the art in RL-based operations research.

The complete modeling framework, instance generator, and documentation will be released as open-source software upon publication, enabling reproducible research and practical deployment in this emerging field.

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A Mathematical Proofs

[Placeholder for mathematical proofs. To be completed if needed.]

A.1 Proof of Rolling Horizon Optimality Conditions

[To be added: Conditions under which rolling horizon policy approximates infinite-horizon optimal policy]

A.2 Proof of State Space Completeness

[To be added: Demonstration that state representation captures all decision-relevant information]

B Parameter Calibration Details

B.1 Complete Parameter Tables

This appendix provides comprehensive parameter tables for all model components, with literature citations and calibration details.

B.1.1 Portfolio Composition Parameters

Table 2: Portfolio Composition Model Parameters

Parameter	Value	Unit	Source
Domestic weight	0.60	fraction	Khanzadi et al. (2018)
International weight	0.40	fraction	Khanzadi et al. (2018)
Portfolio size min	8	projects	Merrow (2011), CII (2019)
Portfolio size max	18	projects	Merrow (2011), CII (2019)

B.1.2 Profit Margin Parameters

Table 3: Profit Margin Distribution Parameters

Parameter	Domestic	International	Source
Mean μ	0.10	0.14	Khanzadi et al. (2018)
Std Dev σ	0.012	0.030	Khanzadi et al. (2018)
Lower bound a	0.08	0.12	Iran Budget Law
Upper bound b	0.12	0.16	Market analysis

B.1.3 Project BAC Parameters

Table 4: Project BAC Lognormal Distribution

Parameter	Value	Source
$\mu_{\log}$	5.01	Merrow (2011)
$\sigma_{\log}$	0.69	Merrow (2011)
Median BAC	\$150M	Merrow (2011)
Mean BAC	\$280M	Merrow (2011)
Min BAC	\$50M	Industry practice
Max BAC	\$5B	Industry practice

B.1.4 S-Curve Parameters

B.1.5 Performance Uncertainty Parameters

B.1.6 Duration Model Parameters

B.1.7 Milestone Payment Structure

B.2 Validation Results

[To be completed: Statistical validation of generated instances against target distributions]

B.2.1 Distributional Validation

[Tables showing sample statistics vs. target values for all key variables]

Table 5: S-Curve Beta Distribution Base Parameters

Parameter	Value	Source
α_{base}	2.5	Cioffi (2005)
β_{base}	2.0	Cioffi (2005)
Peak time τ^*	0.60	Kenley & Wilson (1986)
Size exponent	0.15	Barraza et al. (2007)
Duration exponent	0.12	Barraza et al. (2007)
Residual noise σ	0.10	Calibrated

Table 6: SPI and CPI Distribution Parameters

Parameter	SPI	CPI	Correlation	Source
α	2.8	3.2	–	Merrow (2011)
β	2.2	1.8	–	Love et al. (2016)
Lower bound	0.3	0.5	–	Empirical range
Upper bound	1.3	1.2	–	Empirical range
Mean	0.86	0.87	–	Calibrated
Std Dev	0.18	0.14	–	Calibrated
$\rho(\text{SPI}, \text{CPI})$	–	–	0.68	Love et al. (2016)

B.2.2 Correlation Validation

[Correlation matrices comparing generated data to empirical values]

B.2.3 Portfolio-Level Validation

[Aggregate statistics for generated portfolios]

C Instance Generator Algorithm

C.1 Instance Generator Algorithm

This appendix provides the complete pseudocode for the portfolio instance generator.

Table 7: Project Duration Model Parameters

Parameter	Value	Source
D_{base}	24 months	Merrow (2011)
BAC_{ref}	\$150M	Merrow (2011)
Scale exponent	0.35	Merrow (2011)
Noise σ_D	0.15	Calibrated
Min duration	12 months	Industry practice
Max duration	60 months	Industry practice

Table 8: Standard Milestone Payment Weights

Milestone	Progress Threshold	Payment Weight
Advance	0%	10%
Milestone 1	25%	20%
Milestone 2	50%	30%
Milestone 3	75%	30%
Final	100%	10%

Algorithm 1 Portfolio Instance Generator**Require:** Portfolio size range $[N_{\min}, N_{\max}]$, parameter tables**Ensure:** Portfolio instance $\mathcal{P}$ with all attributes

```

1:
2: Stage 1: Portfolio Composition
3:  $N \leftarrow \text{UniformInt}(N_{\min}, N_{\max})$ 
4:  $N_{\text{domestic}} \leftarrow \lfloor 0.6 \times N \rfloor$ 
5:  $N_{\text{international}} \leftarrow N - N_{\text{domestic}}$ 
6:
7: Stage 2: Project Characteristics
8: for  $i = 1$  to  $N$  do
9:   Assign category: domestic or international
10:  Sample  $\log(\text{BAC}_i) \sim \mathcal{N}(\mu_{\log}, \sigma_{\log})$ 
11:  Clip  $\text{BAC}_i$  to  $[\$50M, \$5B]$ 
12:  Sample  $\pi_i \sim \text{TruncatedNormal}(\mu_{\pi}, \sigma_{\pi}, a_{\pi}, b_{\pi})$ 
13:  Compute  $R_i \leftarrow \text{BAC}_i \times (1 + \pi_i)$ 
14:  Sample  $D_i \leftarrow D_{\text{base}} \times (\text{BAC}_i / \text{BAC}_{\text{ref}})^{0.35} \times \epsilon_D$ 
15:  Clip  $D_i$  to  $[12, 60]$  months
16: end for
17:
18: Stage 3: S-Curve Parameters
19: for  $i = 1$  to  $N$  do
20:   Compute size multiplier:  $m_{\text{size}} \leftarrow (\text{BAC}_i / \text{BAC}_{\text{median}})^{0.15}$ 
21:   Compute duration multiplier:  $m_{\text{dur}} \leftarrow (D_i / D_{\text{median}})^{0.12}$ 
22:   Get category multiplier  $m_{\text{cat}}$  (0.9 for domestic, 1.1 for international)
23:   Sample risk level: low or high
24:   Get risk multipliers  $m_{\text{risk}}^{\alpha}, m_{\text{risk}}^{\beta}$ 
25:   Sample  $\epsilon_{\alpha}, \epsilon_{\beta} \sim \text{LogNormal}(0, 0.10)$ 
26:    $\alpha_i \leftarrow \alpha_{\text{base}} \times m_{\text{size}} \times m_{\text{cat}} \times m_{\text{risk}}^{\alpha} \times \epsilon_{\alpha}$ 
27:    $\beta_i \leftarrow \beta_{\text{base}} \times m_{\text{dur}} \times m_{\text{cat}} \times m_{\text{risk}}^{\beta} \times \epsilon_{\beta}$ 
28:   Clip  $\alpha_i$  to  $[1.2, 5.0]$ ,  $\beta_i$  to  $[1.2, 5.5]$ 
29: end for
30:
31: Stage 4: Cashflow Profiles
32: for  $i = 1$  to  $N$  do
33:   for  $t = 0$  to  $D_i$  do
34:      $\tau \leftarrow t / D_i$ 

```

C.2 Validation Algorithm

Algorithm 2 Instance Validation

Require: Generated portfolio instance $\mathcal{P}$, target statistics

Ensure: Validation report (pass/fail for each check)

```

1:
2: Level 1: Distributional Validation
3: for each variable  $X$  in  $\{\text{BAC}, \pi, D, \alpha, \beta, \text{SPI}, \text{CPI}\}$  do
4:   Compute sample mean  $\bar{X}$ , std dev  $s_X$ 
5:   Check:  $|\bar{X} - \mu_{\text{target}}|/\mu_{\text{target}} < 0.05$  ▷ 5% tolerance
6:   Check:  $|s_X - \sigma_{\text{target}}|/\sigma_{\text{target}} < 0.10$  ▷ 10% tolerance
7: end for
8:
9: Level 2: Correlation Validation
10: Compute  $\rho(\text{SPI}, \text{CPI})$ 
11: Check:  $|\rho - 0.68| < 0.05$ 
12:
13: Level 3: Portfolio-Level Validation
14: Compute total portfolio BAC:  $\sum_{i=1}^N \text{BAC}_i$ 
15: Compute domestic fraction:  $\sum_{i \in \text{domestic}} \text{BAC}_i / \sum_{i=1}^N \text{BAC}_i$ 
16: Check: domestic fraction  $\in [0.55, 0.65]$  ▷ 60%  $\pm$  5%
17:
18: Level 4: Sanity Checks
19: for  $i = 1$  to  $N$  do
20:   Check:  $\text{BAC}_i > 0, D_i > 0, \pi_i > 0$ 
21:   Check:  $\alpha_i \in [1.2, 5.0], \beta_i \in [1.2, 5.5]$ 
22:   Check:  $\text{SPI}_i \in [0.3, 1.3], \text{CPI}_i \in [0.5, 1.2]$ 
23:   Check:  $\sum_m w_m = 1.0$  ▷ Milestone weights sum to 100%
24: end for
25:
26: return Validation report

```

C.3 Implementation Notes

Computational Complexity: The instance generator has time complexity $O(N \times D_{\max})$ where N is portfolio size and $D_{\max}$ is maximum project duration. For typical values ($N = 15, D_{\max} = 60$), generation takes < 1 second per instance.

Reproducibility: All random number generation uses seeded PRNGs (NumPy default_rng with explicit seeds). Seeds are logged for each generated instance.

Parallelization: Instance generation is embarrassingly parallel. For large datasets (10,000+ instances), use multiprocessing with one worker per CPU core.

Output Format: Instances are serialized as JSON or HDF5 files with standardized schema for interoperability.